

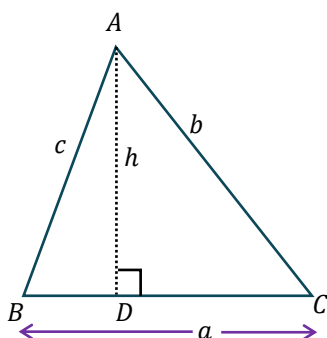
03

Solution of Triangles

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a ΔABC , the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE:



In any triangle ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$$

where R is circumradius and Δ is area of triangle.

Illustration 1

Angles of a triangle are in $4 : 1 : 1$ ratio. The ratio between its greatest side and perimeter is

- (A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Ans. (B)

Solution:

Angles are in ratio $4 : 1 : 1$.

$\Rightarrow$ angles are $120^\circ, 30^\circ, 30^\circ$.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side. Now from sine

formula $\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$

$$\Rightarrow \frac{a}{\sqrt{3}/2} = \frac{b}{1/2} = \frac{c}{1/2}$$

$$\Rightarrow \frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1} = k \text{ (say)}$$

then $a = \sqrt{3}k$, perimeter $= (2 + \sqrt{3})k$

$$\therefore \text{required ratio} = \frac{\sqrt{3}k}{(2 + \sqrt{3})k} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

Illustration 2

In triangle ABC , if $b = 3, c = 4$ and $\angle B = \pi/3$, then number of such triangles is -

- (A) 1 (B) 2 (C) 0 (D) infinite

Ans. (C)

Solution:

Using sine formulae $\frac{\sin B}{b} = \frac{\sin C}{c}$

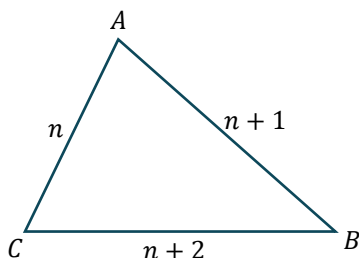
$$\Rightarrow \frac{\sin \pi/3}{3} = \frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6} = \frac{\sin C}{4} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1 \text{ which is not possible.}$$

Hence there exist no triangle with given elements.

Illustration 3

The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.

Solution:



Let the sides be $n, n+1, n+2$ cms.

i.e. $AC = n, AB = n+1, BC = n+2$

Smallest angle is B and largest one is A .

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180^\circ$

$$\Rightarrow 3\angle B + \angle C = 180^\circ \Rightarrow \angle C = 180^\circ - 3\angle B$$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin 3B}{n+1}$$

(i) (ii) (iii)

from (i) and (ii);

$$\frac{2\sin B \cos B}{n+2} = \frac{\sin B}{n} \Rightarrow \cos B = \frac{n+2}{2n} \quad \dots(iv)$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3\sin B - 4\sin^3 B}{n+1} \Rightarrow \frac{\sin B}{n} = \frac{\sin B(3-4\sin^2 B)}{n+1}$$

$$\Rightarrow \frac{n+1}{n} = 3-4(1-\cos^2 B) \quad \dots(v)$$

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from (iv) and (v), we get

$$\begin{aligned}\frac{n+1}{n} &= -1 + 4\left(\frac{n+2}{2n}\right)^2 \Rightarrow \frac{n+1}{n} + 1 = \left(\frac{n^2 + 4n + 4}{n^2}\right) \\ \Rightarrow \frac{2n+1}{n} &= \frac{n^2 + 4n + 4}{n^2} \Rightarrow 2n^2 + n = n^2 + 4n + 4 \\ \Rightarrow n^2 - 3n - 4 &= 0 \Rightarrow (n-4)(n+1) = 0 \\ n &= 4 \text{ or } -1 \\ \text{where } n &\neq -1 \\ \therefore n &= 4. \text{ Hence the sides are } 4, 5, 6\end{aligned}$$

2. COSINE FORMULAE:

$$(a) \cos A = \frac{b^2 + c^2 - a^2}{2bc} \text{ or } a^2 = b^2 + c^2 - 2bc \cos A$$

$$(b) \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$(c) \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Illustration 4

In a triangle ABC , if $B = 30^\circ$ and $c = \sqrt{3}b$, then A can be equal to -

- (A) 45° (B) 60° (C) 90° (D) 120°

Ans. (C)

Solution:

$$\text{We have } \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{3b^2 + a^2 - b^2}{2 \times \sqrt{3}b \times a}$$

$$\Rightarrow a^2 - 3ab + 2b^2 = 0$$

$$\Rightarrow (a - 2b)(a - b) = 0$$

$$\Rightarrow \text{Either } a = b \Rightarrow A = 30^\circ$$

$$\text{or } a = 2b \Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90^\circ.$$

Illustration 5

In a triangle ABC , $(a^2 - b^2 - c^2) \tan A + (a^2 - b^2 + c^2) \tan B$ is equal to -

- (A) $(a^2 + b^2 - c^2) \tan C$ (B) $(a^2 + b^2 + c^2) \tan C$
(C) $(b^2 + c^2 - a^2) \tan C$ (D) none of these

Ans. (D)

Solution:

Using cosine law :

The given expression is equal to $-2bc \cos A \tan A + 2ac \cos B \tan B$

$$= 2abc \left(-\frac{\sin A}{a} + \frac{\sin B}{b} \right) = 0$$

Illustration 6

In a triangle ABC , A, B, C are in A.P. Show that $2\cos\left(\frac{A-C}{2}\right) = \frac{a+c}{\sqrt{a^2-ac+c^2}}$.

Solution:

$$A + C = 2B \Rightarrow A + B + C = 3B \Rightarrow B = 60^\circ$$

$$\therefore \cos 60^\circ = \frac{a^2+c^2-b^2}{2ac} \Rightarrow a^2-ac+c^2=b^2$$

$$\begin{aligned} \Rightarrow \frac{a+c}{\sqrt{a^2-ac+c^2}} &= \frac{a+c}{b} = \left[\frac{\sin A + \sin C}{\sin B} \right] = \frac{2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)}{\sin B} \\ &= 2\cos\frac{A-C}{2} \quad (\because A+C=2B) \end{aligned}$$

Illustration 7

If in a triangle ABC , $\frac{\cos A + 2\cos C}{\cos A + 2\cos B} = \frac{\sin B}{\sin C}$ prove that the triangle ABC is either isosceles or right angled.

Solution:

$$\begin{aligned} &\cos A(\sin B - \sin C) + (\sin 2B - \sin 2C) = 0 \\ \Rightarrow &\cos A(\sin B - \sin C) + 2\cos(B+C)\sin(B-C) = 0 \quad \because B+C=\pi-A \\ \Rightarrow &\cos A(\sin B - \sin C) - 2\cos A\sin(B-C) = 0 \\ \Rightarrow &\cos A[(\sin B - \sin C) - 2(\sin B \cos C - \cos B \sin C)] = 0 \\ \Rightarrow &\text{either } \cos A = 0 \Rightarrow A = 90^\circ \Rightarrow \text{right angled} \\ \text{or } &(\sin B - \sin C) - 2(\sin B \cos C - \cos B \sin C) = 0 \\ \Rightarrow &(b-c) - 2\left(b \cdot \frac{a^2+b^2-c^2}{2ab} - c \cdot \frac{a^2+c^2-b^2}{2ac}\right) = 0 \\ \Rightarrow &a(b-c) - 2(b^2 - c^2) = 0 \\ (b-c)[a - 2(b+c)] &= 0 \\ \therefore b-c=0 \Rightarrow b=c &\Rightarrow \text{isosceles} \end{aligned}$$

3. PROJECTION FORMULAE:

(a) $b \cos C + c \cos B = a$

(b) $c \cos A + a \cos C = b$

(c) $a \cos B + b \cos A = c$

Illustration 8

In a ΔABC , $c \cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution:

$$\text{Here, } \frac{c}{2}(1+\cos A) + \frac{a}{2}(1+\cos C) = \frac{3b}{2}$$

$$\Rightarrow a+c+(c \cos A + a \cos C) = 3b$$

$$\Rightarrow a+c+b = 3b \quad \{\text{using projection formula}\}$$

$$\Rightarrow a+c = 2b$$

which shows a, b, c are in A.P.

Solution of Triangles

4. NAPIER'S ANALOGY (TANGENT RULE):

$$(a) \tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$(b) \tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2}$$

$$(c) \tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}$$

Illustration 9

In a $\triangle ABC$, the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

Solution:

$$\text{Here, } \tan\left(\frac{A-B}{2}\right) = \frac{1}{3} \tan\left(\frac{A+B}{2}\right) \quad \dots(i)$$

$$\text{using Napier's analogy, } \tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cdot \cot\left(\frac{C}{2}\right) \quad \dots(ii)$$

from (i) & (ii) ;

$$\frac{1}{3} \tan\left(\frac{A+B}{2}\right) = \frac{a-b}{a+b} \cdot \cot\left(\frac{C}{2}\right) \Rightarrow \frac{1}{3} \cot\left(\frac{C}{2}\right) = \frac{a-b}{a+b} \cdot \cot\left(\frac{C}{2}\right)$$

$$\{ \text{as } A+B+C=\pi \quad \therefore \tan\left(\frac{B+C}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cot \frac{C}{2} \}$$

$$\Rightarrow \frac{a-b}{a+b} = \frac{1}{3} \quad \text{or} \quad 3a - 3b = a + b$$

$$2a = 4b \quad \text{or} \quad \frac{a}{b} = \frac{2}{1} \Rightarrow \frac{b}{a} = \frac{1}{2}$$

Thus the ratio of the sides opposite to the angles is $b : a = 1 : 2$.

Ans.

5. HALF ANGLE FORMULAE:

$$s = \frac{a+b+c}{2} = \text{semi-perimeter of triangle.}$$

$$(a) (i) \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}} \quad (ii) \sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}} \quad (iii) \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$(b) (i) \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}} \quad (ii) \cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}} \quad (iii) \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

$$(c) (i) \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \quad (ii) \tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}} \quad (iii) \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$= \frac{\Delta}{s(s-a)} \quad = \frac{\Delta}{s(s-b)} \quad = \frac{\Delta}{s(s-c)}$$

(d) Area of Triangle

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3,$$

where p_1, p_2, p_3 are altitudes from vertices A, B, C respectively.

Illustration 10

If in a triangle ABC , CD is the angle bisector of the angle ACB , then CD is equal to-

- (A) $\frac{a+b}{2ab} \cos \frac{C}{2}$ (B) $\frac{2ab}{a+b} \sin \frac{C}{2}$ (C) $\frac{2ab}{a+b} \cos \frac{C}{2}$ (D) $\frac{b \sin \angle DAC}{\sin(B+C/2)}$

Ans. (C, D)

Solution:

$$\Delta CAB = \Delta CAD + \Delta CDB$$

$$\Rightarrow \frac{1}{2} ab \sin C = \frac{1}{2} b \cdot CD \cdot \sin\left(\frac{C}{2}\right) + \frac{1}{2} a \cdot CD \sin\left(\frac{C}{2}\right)$$

$$\Rightarrow CD(a+b) \sin\left(\frac{C}{2}\right) = ab \left(2 \sin\left(\frac{C}{2}\right) \cos\left(\frac{C}{2}\right)\right)$$

$$\text{So } CD = \frac{2ab \cos(C/2)}{(a+b)}$$

$$\text{and in } \Delta CAD, \frac{CD}{\sin \angle DAC} = \frac{b}{\sin \angle CDA} \quad (\text{by sine rule})$$

$$\Rightarrow CD = \frac{b \sin \angle DAC}{\sin(B+C/2)}$$

Illustration 11

If Δ is the area and $2s$ the sum of the sides of a triangle, then show $\Delta \leq \frac{s^2}{3\sqrt{3}}$.

Solution:

$$\text{We have, } 2s = a + b + c, \Delta^2 = s(s-a)(s-b)(s-c)$$

Now, A.M. $\geq$ G.M.

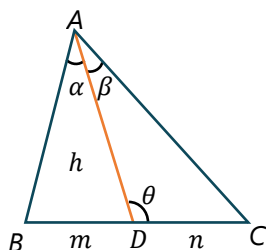
$$\frac{(s-a) + (s-b) + (s-c)}{3} \geq \{(s-a)(s-b)(s-c)\}^{1/3}$$

$$\text{or } \frac{3s-2s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{\Delta^2}{s} \leq \frac{s^3}{27} \Rightarrow \Delta \leq \frac{s^2}{3\sqrt{3}} \quad \text{Ans}$$

6. $m-n$ THEOREM:



$$(m+n) \cot \theta = m \cot \alpha - n \cot \beta$$

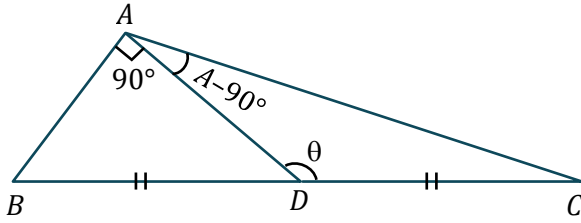
$$(m+n) \cot \theta = n \cot B - m \cot C.$$

Illustration 12

In a $\triangle ABC$, AD divides BC in the ratio $2 : 1$ such that at $\angle BAD = 90^\circ$ then prove that $\tan A + 3 \tan B = 0$

Solution:

From the figure, we see that $\theta = 90^\circ + B$ (as θ is external angle of $\triangle ABD$)



Now if we apply **m-n rule** in $\triangle ABC$, we get
 $(2 + 1) \cot (90^\circ + B) = 2 \cdot \cot 90^\circ - 1 \cdot \cot (A - 90^\circ)$
 $\Rightarrow -3 \tan B = \cot (90^\circ - A)$
 $\Rightarrow -3 \tan B = \tan A$
 $\Rightarrow \tan A + 3 \tan B = 0$

Illustration 13

The base of a $\triangle$ is divided into three equal parts. If α, β, γ be the angles subtended by these parts at the vertex, prove that :

$$(\cot \alpha + \cot \beta) (\cot \beta + \cot \gamma) = 4 \operatorname{cosec}^2 \beta$$

Solution:

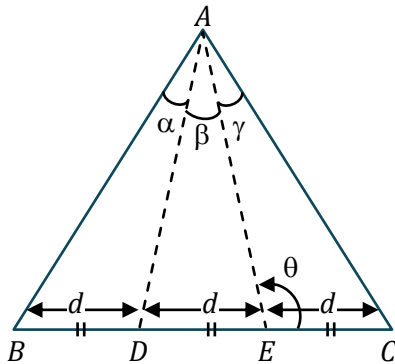
Let point D and E divide the base BC into three equal parts i.e. $BD = DE = EC = d$ (Let) and let α, β and γ be the angles subtended by BD, DE and EC respectively at their opposite vertex. Now in $\triangle ABC$

$$\therefore BE : EC = 2d : d = 2 : 1$$

$\therefore$ from **m-n rule**, we get

$$(2 + 1) \cot \theta = 2 \cot (\alpha + \beta) - \cot \gamma$$

$$\Rightarrow 3 \cot \theta = 2 \cot (\alpha + \beta) - \cot \gamma \quad \dots(i)$$



again

$\therefore$ in $\triangle ADC$

$$\therefore DE : EC = d : d = 1 : 1$$

$\therefore$ if we apply **m-n rule** in $\triangle ADC$, we get

$$(1 + 1) \cot \theta = 1 \cdot \cot \beta - 1 \cdot \cot \gamma$$

$$2 \cot \theta = \cot \beta - \cot \gamma \quad \dots(ii)$$

$$\text{from (i) and (ii), we get } \frac{3 \cot \theta}{2 \cot \theta} = \frac{2 \cot (\alpha + \beta) - \cot \gamma}{\cot \beta - \cot \gamma}$$

$$\Rightarrow 3 \cot \beta - 3 \cot \gamma = 4 \cot (\alpha + \beta) - 2 \cot \gamma$$

$$\Rightarrow 3 \cot \beta - \cot \gamma = 4 \cot (\alpha + \beta)$$

$$\Rightarrow 3\cot\beta - \cot\gamma = 4 \left\{ \frac{\cot\alpha \cdot \cot\beta - 1}{\cot\beta + \cot\alpha} \right\}$$

$$\Rightarrow 3\cot^2\beta + 3\cot\alpha \cot\beta - \cot\beta \cot\gamma - \cot\alpha \cot\gamma = 4 \cot\alpha \cot\beta - 4$$

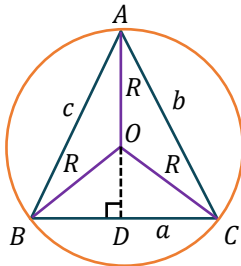
$$\Rightarrow 4 + 3\cot^2\beta = \cot\alpha \cot\beta + \cot\beta \cot\gamma + \cot\alpha \cot\gamma$$

$$\Rightarrow 4 + 4\cot^2\beta = \cot\alpha \cot\beta + \cot\alpha \cot\gamma + \cot\beta \cot\gamma + \cot^2\beta$$

$$\Rightarrow 4(1 + \cot^2\beta) = (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma)$$

$$\Rightarrow (\cot\alpha + \cot\beta)(\cot\beta + \cot\gamma) = 4\operatorname{cosec}^2\beta$$

7. RADIUS OF THE CIRCUMCIRCLE 'R':



Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

Illustration 14

In a $\triangle ABC$, prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

Solution:

$$\text{In a } \triangle ABC, \text{ we know that } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{and } \sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$$

$$= \frac{4abc}{8R^3} = \frac{16\Delta R}{8R^3} = \frac{2\Delta}{R^2}$$

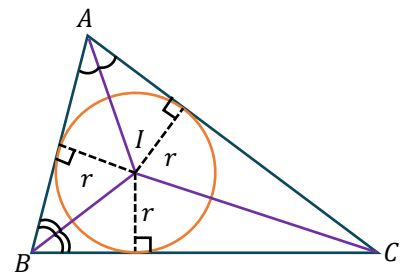
8. RADIUS OF THE INCIRCLE 'r':

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.

$$r = \frac{\Delta}{s} = (s-a)\tan \frac{A}{2} = (s-b)\tan \frac{B}{2}$$

$$= (s-c)\tan \frac{C}{2} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$

$$= a \frac{\sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = b \frac{\sin \frac{A}{2} \sin \frac{C}{2}}{\cos \frac{B}{2}} = c \frac{\sin \frac{B}{2} \sin \frac{A}{2}}{\cos \frac{C}{2}}$$



Solution of Triangles

Illustration 15

In a triangle ABC , if $a : b : c = 4 : 5 : 6$, then ratio between its circumradius and inradius is-

- (A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Ans. (A)

Solution:

$$\frac{R}{r} = \frac{abc}{4\Delta} \cdot \frac{\Delta}{s} = \frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)} \quad \dots(i)$$

$$\because a : b : c = 4 : 5 : 6 \Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (say)}$$

$$\Rightarrow a = 4k, b = 5k, c = 6k$$

$$\therefore s = \frac{a+b+c}{2} = \frac{15k}{2}, s-a = \frac{7k}{2}, s-b = \frac{5k}{2}, s-c = \frac{3k}{2}$$

$$\text{using (i) in these values } \frac{R}{r} = \frac{(4k)(5k)(6k)}{4 \left(\frac{7k}{2}\right) \left(\frac{5k}{2}\right) \left(\frac{3k}{2}\right)} = \frac{16}{7}$$

Illustration 16

If A, B, C are the angles of a triangle, prove that : $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.

Solution:

$$\begin{aligned} \cos A + \cos B + \cos C &= 2 \cos \left(\frac{A+B}{2} \right) \cdot \cos \left(\frac{A-B}{2} \right) + \cos C \\ &= 2 \sin \frac{C}{2} \cdot \cos \left(\frac{A-B}{2} \right) + 1 - 2 \sin^2 \frac{C}{2} = 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \sin \left(\frac{C}{2} \right) \right] \\ &= 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right] \quad \left\{ \because \frac{C}{2} = 90^\circ - \left(\frac{A+B}{2} \right) \right\} \\ &= 1 + 2 \sin \frac{C}{2} \cdot 2 \sin \frac{A}{2} \cdot \sin \frac{B}{2} = 1 + 4 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} \\ &= 1 + \frac{r}{R} \quad \{ \text{as, } r = 4R \sin A/2 \cdot \sin B/2 \cdot \sin C/2 \} \end{aligned}$$

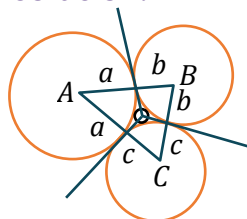
$$\Rightarrow \cos A + \cos B + \cos C = 1 + \frac{r}{R}. \text{ Hence proved.}$$

Illustration 17

Three circles, whose radii are a, b and c , touch one another externally and the tangents at their points of contact meet in a point, prove that the distance of this point from either of their points of contact is

$$\left(\frac{abc}{a+b+c} \right)^{\frac{1}{2}}.$$

Solution:



required distance = inradius of $\triangle ABC$

$$\begin{aligned}\therefore 2s &= a + b + b + c + c + a \\ &= 2(a + b + c)\end{aligned}$$

$$s = a + b + c$$

$$\Delta = \sqrt{s(s-(a+b))(s-(b+c))(s-(c+a))} = \sqrt{(a+b+c)(abc)}$$

$\therefore$ required distance

$$= \frac{\Delta}{s} = \frac{\sqrt{(a+b+c)(abc)}}{(a+b+c)} = \sqrt{\frac{abc}{a+b+c}} = \left(\frac{abc}{a+b+c}\right)^{\frac{1}{2}}$$

9. RADIUS OF THE EX- CIRCLES:

If r_1, r_2, r_3 are the radii of the ex-circles of $\triangle ABC$ opposite to the vertex A, B, C respectively, then

$$(i) \quad r_1 = \frac{\Delta}{s-a}; r_2 = \frac{\Delta}{s-b}; r_3 = \frac{\Delta}{s-c};$$

$$(ii) \quad r_1 = s \tan \frac{A}{2}; r_2 = s \tan \frac{B}{2}; r_3 = s \tan \frac{C}{2}$$

$$(iii) \quad r_1 = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}} \text{ and so on}$$

$$(iv) \quad r_1 = 4R \sin \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2} \text{ and so on}$$

Illustration 18

In a triangle ABC , find the value of $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3}$.

Solution:

$$\begin{aligned}& \left(\frac{b-c}{\frac{\Delta}{s-a}}\right) + \left(\frac{c-a}{\frac{\Delta}{s-b}}\right) + \left(\frac{a-b}{\frac{\Delta}{s-c}}\right) \\ &= \frac{1}{\Delta} [(b-c)(s-a) + (c-a)(s-b) + (a-b)(s-c)] \\ &= \frac{1}{\Delta} [s(b-c+c-a+a-b) - a(b-c) - b(c-a) - c(a-b)] = 0\end{aligned}$$

Illustration 19

In an acute angled triangle ABC , $r + r_1 = r_2 + r_3$ and $\angle B > \frac{\pi}{3}$, then prove that $b + 3c < 3a < 3b + 3c$

Solution:

$$r - r_2 = r_3 - r_1 \Rightarrow \frac{\Delta}{s} - \frac{\Delta}{s-b} = \frac{\Delta}{s-c} - \frac{\Delta}{s-a}$$

$$\text{or } \frac{-b}{s(s-b)} = \frac{c-a}{(s-c)(s-a)}$$

$$\text{or } \frac{(s-a)(s-c)}{s(s-b)} = \frac{a-c}{b} \Rightarrow \tan^2 \frac{B}{2} = \frac{a-c}{b}$$

$$\text{But } \frac{B}{2} \in \left(\frac{\pi}{6}, \frac{\pi}{4}\right). \text{ Therefore,}$$

$$\tan^2 \frac{B}{2} \in \left(\frac{1}{3}, 1\right) \Rightarrow \frac{1}{3} < \frac{a-c}{b} < 1$$

$$\text{or } b < 3a - 3c < 3b$$

$$b + 3c < 3a < 3b + 3c$$

10. ANGLE BISECTORS & MEDIANS AND ALTITUDE:

- (i) An angle bisector divides the base in the ratio of corresponding sides.

$$\frac{BD}{CD} = \frac{c}{b} \Rightarrow BD = \frac{ac}{b+c} \text{ \& } CD = \frac{ab}{b+c}$$

- (ii) If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} \text{ and } \beta_a = \frac{2bc \cos \frac{A}{2}}{b+c}$$

$$\text{Note that } m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$$

- (iii) Length of altitude from the angle $A = A_a = \frac{2\Delta}{a}$

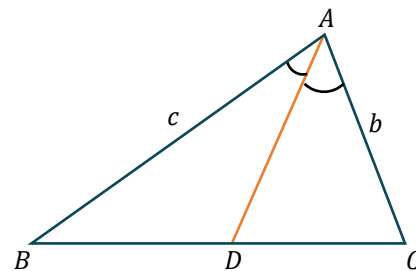


Illustration 20

In $\triangle ABC$, AD & BE are its two median. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$ then find the length of BE and area of $\triangle ABC$.

Solution:

$$AP = \frac{2}{3}; AD = \frac{8}{3}; PD = \frac{4}{3}; \text{ Let } PB = x$$

$$\tan 60^\circ = \frac{8/3}{x} \text{ or } x = \frac{8}{3\sqrt{3}}$$

$$\text{Area of } \triangle ABP = \frac{1}{2} \times \frac{8}{3} \times \frac{8}{3\sqrt{3}} = \frac{32}{9\sqrt{3}}$$

$$\therefore \text{Area of } \triangle ABC = 3 \times \frac{32}{9\sqrt{3}} = \frac{32}{3\sqrt{3}}$$

$$\text{Also, } BE = \frac{3}{2}x = \frac{4}{\sqrt{3}}$$

Illustration 21

If AD , BE and CF are the medians of a $\triangle ABC$, then $(AD^2 + BE^2 + CF^2) : (BC^2 + CA^2 + AB^2)$ is equal to
 (A) 4 : 3 (B) 3 : 2 (C) 3 : 4 (D) 2 : 3

Ans. (C)

Solution:

$$\therefore AD^2 = \frac{1}{4} (2b^2 + 2c^2 - a^2),$$

$$BE^2 = \frac{1}{4} (2c^2 + 2a^2 - b^2) \text{ and}$$

$$CF^2 = \frac{1}{4} (2a^2 + 2b^2 - c^2) \quad \therefore \frac{AD^2 + BE^2 + CF^2}{BC^2 + CA^2 + AB^2} = \frac{3}{4}$$

Illustration 22

If p_1, p_2, p_3 are respectively the lengths of perpendiculars from the vertices of a triangle ABC to the opposite sides, prove that :

$$(i) \frac{\cos A}{p_1} + \frac{\cos B}{p_2} + \frac{\cos C}{p_3} = \frac{1}{R} \quad (ii) \frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b} = \frac{a^2 + b^2 + c^2}{2R}$$

Solution:

$$(i) \text{ use } \frac{1}{p_1} = \frac{a}{2\Delta}, \frac{1}{p_2} = \frac{b}{2\Delta}, \frac{1}{p_3} = \frac{c}{2\Delta}$$

$$\therefore \text{LHS} = \frac{1}{2\Delta} (a \cos A + b \cos B + c \cos C)$$

$$= \frac{R}{2\Delta} (\sin 2A + \sin 2B + \sin 2C) = \frac{4R \sin A \sin B \sin C}{2\Delta}$$

$$= \frac{4R}{2\Delta} \cdot \frac{a}{2R} \cdot \frac{b}{2R} \cdot \frac{c}{2R} = \frac{1}{4\Delta R^2} abc = \frac{1}{4\Delta R^2} \cdot (4\Delta R) = \frac{1}{R} = \text{RHS}$$

$$(ii) \text{LHS} = \frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b} = \frac{a^2 + b^2 + c^2}{2R} = \frac{2b\Delta}{ac} + \frac{2c\Delta}{ab} + \frac{2a\Delta}{bc} = \frac{2\Delta(a^2 + b^2 + c^2)}{abc}$$

$$= \frac{2\Delta(a^2 + b^2 + c^2)}{4\Delta R} = \frac{a^2 + b^2 + c^2}{2R}$$

11. AMBIGUOUS CASE:

When two sides and an angle opposite to one of them is given (Ambiguous case):

$$\text{By applying cosine rule, we have } \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\Rightarrow a^2 - (2c \cos B)a + (c^2 - b^2) = 0 \Rightarrow a = c \cos B \pm \sqrt{(c \cos B)^2 - (c^2 - b^2)}$$

$$\Rightarrow a = c \cos B \pm \sqrt{b^2 - (c \sin B)^2}$$

This equation leads to following cases :

Case-I : If $b < c \sin B$, no such triangle is possible.

Case-II : Let $b = c \sin B$. There are further following case :

(a) B is an obtuse angle $\Rightarrow \cos B$ is negative. There exists no such triangle.

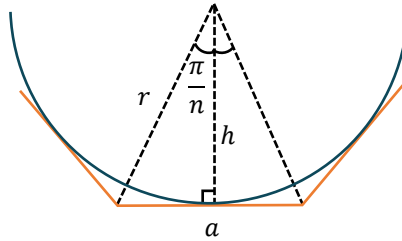
(b) B is an acute angle $\Rightarrow \cos B$ is positive. There exists only one such triangle.

Case-III : Let $b > c \sin B$. There are further following cases :

(a) B is an acute angle $\Rightarrow \cos B$ is positive. In this case triangle will exists if and only if $c \cos B \sqrt{b^2 - (c \sin B)^2} > \text{or } c > b \Rightarrow$ Two such triangle is possible. If $c < b$, only one such triangle is possible.

(b) B is an obtuse angle $\Rightarrow \cos B$ is negative. In this case two values of a will exist if and only if $\sqrt{b^2 - (c \sin B)^2} > |c \cos B| \Rightarrow b > c$. So in this case only one such triangle is possible. If $b < c$ there exists no such triangle.

12. REGULAR POLYGON:



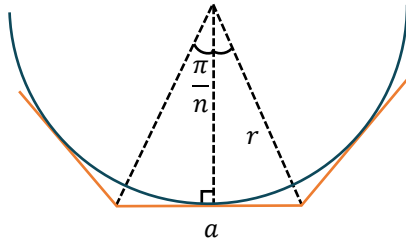
A regular polygon has all its sides equal. It may be inscribed or circumscribed.

(a) Inscribed in Circle of Radius r :

$$(i) \quad a = 2h \tan \frac{\pi}{n} = 2r \sin \frac{\pi}{n}$$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr \sin \frac{\pi}{n}$ and $A = \frac{1}{2} nr^2 \sin \frac{2\pi}{n}$

(b) Circumscribed About a Circle of Radius r :



$$(i) \quad a = 2r \tan \frac{\pi}{n}$$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $p = 2nr \tan \frac{\pi}{n}$ and $A = nr^2 \tan \frac{\pi}{n}$

Illustration 23

A regular hexagon & a regular dodecagon are inscribed in the same circle. If the side of the dodecagon is $(\sqrt{3} - 1)$, if the side of the hexagon is $\sqrt[4]{k}$, then find value of k .

Ans. $\sqrt{2}$

Solution:

$$\text{For dodecagon } \angle A'OB' = \frac{2\pi}{12} = 30^\circ$$

$$\Rightarrow \angle OA'B' = \angle OB'A' = 75^\circ \Rightarrow \frac{R}{\sin 75^\circ} = \frac{\sqrt{3}-1}{\sin 30^\circ}$$

$$\Rightarrow R = \frac{(\sqrt{3}-1)(\sqrt{3}+1)}{2\sqrt{2} \times \frac{1}{2}} \Rightarrow R = \sqrt{2}$$

$$\text{For hexagon } \angle AOB = \frac{2\pi}{6} = 60^\circ$$

$$\Rightarrow \triangle AOB \text{ is equilateral } \Rightarrow AB = R = \sqrt{2}$$

